



A Physics-Informed Neural-Network **Digital Twin** of a Four-Mecanum-Wheel Omnidirectional Platform

High-fidelity-simulation accounting for slip and spin with LuGre-Hertzian friction model, adaptive sliding-mode control, and data generation with holistic excitation profiles for **forward-dynamics learning** and **inverse friction force identification**.

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4 Mecanum-Wheeled Omnidirectional Mobile Robot · slip-spin digital twin

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Outline

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Motivation & research relevance

Market & applications · the control-fidelity gap · related work & limitations · PINN-based solution & safety certifiability

I

The plant — youBot Mecanum platform

Geometry & parameters · O vs X roller configuration · no-slip kinematics & Jacobian

II

Roller slip & spin in the model

Discrete rollers (12/wheel) · contact hand-off · passive roller DOF

III

Composite LuGre–Adamov friction

The pre-slip problem · bristle dynamics & slip–spin coupling

IV

Adaptive sliding-mode control & stability

Velocity vs position references · uniform ultimate boundedness

V

Excitation design, diagnostics & data pipeline

Friction-circle caps · seven trajectories · solver selection · plant → PINN · diagnostic screening & whitelist (5345/5670)

VI

Identifiability of variables & forces

χ spin-gated & μ slip-gated from the observable forces · the thin identifiable tail · roller observability & force anisotropy · force-law adequacy for the PINN head

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CHAPTER 1 · MOTIVATION

Mecanum robot market & applications

Market growth, key sectors, and the deployment barriers that gate real-world use.

1.1 Market overview and growth projections

1.2 Application sectors and recurring barriers

Mecanum-wheeled robot market overview

Rapid, autonomy-concentrated growth — 2024 baselines against 2030–2035 targets

Market segment	2024	Target	CAGR
Mecanum wheel systems	\$1.17 B	\$2.71 B (2033)	9.6%
Total mobile robots	~\$5 B	\$14 B (2030)	19.0%
AGV–AMR (combined)	\$5.66 B	\$20 B (2035)	12.1%
AMR (sub-segment)	\$2.77 B	\$10.36 B (2034)	15.8%

Key drivers: AGV/AMR proliferation, warehouse automation, Industry 4.0, e-commerce and labour shortages, AI navigation.

WHY THIS MATTERS FOR CONTROL

The Mecanum-wheel market grows at **9.6% CAGR to \$2.71 B by 2033**; the broader mobile-robot market expands **~2× faster (19%)**. Growth is concentrated in **autonomy and control**, not the wheel itself.

At this scale, control-fidelity problems — slip, energy loss, odometry drift — become **economically significant**, not academic curiosities.

Application sectors & recurring barriers

DEPLOYMENT SECTORS

Warehouse & intralogistics

Transport, picking, sorting

Manufacturing & assembly

Feeding, material handling

Hospital & clinical logistics

Medication, sample delivery

Cleaning & inspection

Maintenance, surveillance

Assistive mobility

Wheelchairs, personal transport

FOUR RECURRING BARRIERS — BURKACKI ET AL. (2025)

Limited stability on uneven terrain

Maneuverability degrades on inclined and uneven surfaces where roller-ground contact becomes unpredictable.

High energy consumption

Wheel slippage causes measurable energy loss (6.18% documented), cutting battery-supported range proportionally.

Complex control requirements

Kinematic models omit nonlinear, load- and surface-dependent friction that pure controllers cannot reject.

Elevated cost

Precision requires external metrology (optical tracking, mechanical guides) rather than onboard control fidelity.

The barriers split cleanly: **control-related** (slip, odometry drift, disturbance rejection) versus **non-control** (cost, battery, integration, standards). **This work targets the control-related half.**

[1] M. Burkacki et al., "Systematic review of Mecanum and Omni-wheel technologies for motor impairments," *Applied Sciences* 15(9):4773, 2025.

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CHAPTER TWO · MOTIVATION

The control-fidelity gap

Physics-level obstacles the ideal-rolling model omits — and the failures they cause.

2.1 Why higher-fidelity control is needed

2.2 Sector barriers and where control is decisive

Physics-level obstacles in real-world deployment

The ideal-rolling kinematic model omits these — each has caused an independently reported failure

ENERGY LOSS FROM WHEEL SLIPPAGE — ~6.18%

A real industrial haul (5 tonnes of zinc ingots, 5 km) lost ~6.18% of energy to slippage, cutting battery range proportionally. A kinematic controller cannot account for it. [Bayar et al., 2025]

ORIENTATION DEVIATIONS IN TRACKING

Roller-surface contact produces significant orientation deviations during tracking, needing a ramp-shaped reference intervention beyond baseline tracking to suppress. [Ortiz Hernández & Rosas Almeida, 2024]

SURFACE- & VELOCITY-DEPENDENT ODOMETRY DRIFT

Odometric drift varies with floor surface and speed — the same controller yields different positioning error across floors and speeds, undermining pose repeatability. [Shahidi et al., 2021]

COUPLED ACCURACY / ENERGY DEGRADATION

Restricted force/torque authority causes transient overshoot that simultaneously lowers tracking and raises energy use, reducing autonomy. [Ramírez-Neria et al., 2025]

ROOT CAUSE — UNDERMODELED FRICTION

Mecanum friction is comparatively undermodeled. An experimentally validated **orthotropic-friction** model shows contact behaviour depends on **both velocity and the payload-shifted centre of mass** [Manzl et al., 2024]. These effects are nonlinear, load- and surface-dependent — kinematic controllers cannot reject them.

Sector-specific control barriers

Sector	Control barrier	Evidence / impact
Heavy industry	Slip-driven energy loss shrinks range	5-t zinc haul: 6.18% loss — Bayar et al. 2025 [2]
Automotive intralogistics	Baseline controllers lack robustness	Adaptive SMC + ESO — Bui et al. 2025 [10]
Warehouse logistics	Odometry drift breaks repeatability	Surface/velocity error — Shahidi et al. 2021 [4]
Aerospace assembly	0.01 mm tolerance vs. low accuracy	Two-stage: wheels + laser — Zhao et al. 2026 [11]
Hospitals	6 risk dimensions, preliminary testing	Hygiene, interaction — Bernhard et al. 2024 [12]

BENCHMARK — KUKA OMNIMOVE



Key insight: deployed precision is bought by **external referencing**, not onboard control fidelity. Mecanum slip / odometry drift forces this dependence — the problem this work addresses.

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CHAPTER THREE · MOTIVATION

Related work & limitations

Higher-fidelity control closes measurable gaps — yet none recovers friction online.

3.1 Higher-fidelity model-based control results

3.2 Deployed control vs. the research frontier

Higher-fidelity model-based control: measurable gains

Gap addressed	Approach	Reported gain	Reference
Model fidelity / tracking error	Complete Lagrangian + linearized motor	17% lower DTW error	Li & Li 2025 [7]
Disturbance rejection (load, noise)	IMC + disturbance observer	97.6% / 98.3% faster settling	Li & Li 2025 [7]
Inclined surfaces (slip, gravity)	ADRC	49–60% better tracking	Ortiz Hernández 2025 [21]
Unknown CoG offset + slope	MRAC, Lyapunov adaptation	Real-time comp.; stability proof	Chaichumporn 2025 [22]
Disturbance + model uncertainty	AVPSMO feedforward in MPC	Lyapunov-proven suppression	ISA Trans. 2024 [8]

All figures are the cited authors' own results against their own baselines.

THE COMMON THREAD

Improving model fidelity and adding explicit disturbance estimation **measurably** improves tracking and robustness. The residual physics — **slip, friction, load** — must be modeled or estimated, not ignored.

THE REMAINING GAP

Yet **none recovers per-wheel friction parameters online**, and none is standard deployed practice. This is the gap the PINN + Mamba digital twin targets.

Deployed control vs. the research frontier

WHAT INDUSTRY SHIPS

- **Standard architecture:** kinematic inverse-Mecanum + per-wheel PID beneath SLAM / navigation.
- **Accuracy secured by:** mechanical precision + external referencing (optical tracking, guides).
- **KUKA omniMove:** base drive ± 5 mm; contactless stated ± 3 mm; ± 1 mm only with optical / mechanical aids.

WHAT THE LITERATURE PROPOSES

- Adaptive integral terminal sliding mode [Sun et al., 2021].
- Anti-peaking linear ADRC [Ramírez-Neria et al., 2025].
- Sliding-mode-observer MPC [ISA Trans., 2024]; adaptive SMC + ESO [Bui et al., 2025].
- **Repeatedly show:** kinematic controllers are insufficient under real disturbance — yet none is standard practice.

THE OPENING

Close the **fidelity / identifiability gap** between deployed kinematic control and the slip/friction reality the frontier documents. **No prior work recovers per-wheel friction (μ , χ) online for Mecanum.**

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CHAPTER FOUR · MOTIVATION

PINN-based solution & its relevance

Physics-informed learning for friction and dynamics — and why interpretability enables deployment.

4.1 Physics-informed learning for friction & dynamics

4.2 Safety certifiability & the gap this work addresses

Physics-informed learning for friction & dynamics

LUGRE FRICTION-PARAMETER ID

A PINN fuses the LuGre dynamic friction model with learnable neural components to identify LuGre parameters from small, noisy datasets at higher fidelity than simplified models. [Ozmen et al., 2026, ACC]

JOINT DYNAMICS + RUNGE-KUTTA CELLS

A PINN encoding joint dynamics inside a recurrent net with custom RK cells outperforms conventional PINNs and grey-box SSID, matching ground-truth parameters on UR3e hardware. [Yang et al., 2023, RA-L]

ENHANCED PINNS — MOTOR & FORCE COUPLING

+31–37% modeling-accuracy gain and –40–52% tracking-error reduction over prior SOTA. [FITEE, 2025]

NONSMOOTH MULTI-BODY PINNS (STICK-SLIP)

PINNs embedding nonsmooth multi-body formulations capture stick-slip and separation-reattachment without the vanishingly small time steps of conventional time-stepping. [Li et al., 2024, Nonlinear Dyn.]

KAN FOR STATIC FRICTION

Kolmogorov–Arnold-network methods recover interpretable closed-form static-friction laws ($R^2 > 0.95$) with no predefined functional form — directly serving the interpretability requirement for certification. [Wang et al., 2025]

CRITICAL CAVEAT

Rigid-body-prior networks alone are insufficient: standard Deep Lagrangian Networks **fail** to capture nonlinear friction and flexibility, motivating augmentations (DeLaN-FFNN). This justifies pairing a physics prior **with** a learnable friction term. [Li et al., 2025]

Safety certifiability & the gap this work addresses

WHY INTERPRETABILITY MATTERS FOR DEPLOYMENT

ISO 3691-4:2023 mandates **PL d (SIL 2)** for driverless industrial trucks in shared workspaces. Black-box neural controllers are **confined to the lowest safety-integrity levels** — outputs cannot be exhaustively bounded or formally verified [Zhang & Li, 2020; Tambon et al., 2022]. A physics-informed model with known friction structure (LuGre) and interpretable per-wheel parameters **enables the verification evidence black boxes cannot supply.**

TWO OBSERVATIONS DEFINING THE OPENING

- (1) The strongest PINN friction-ID and forward-dynamics evidence comes from **fixed-base manipulators**, not Mecanum platforms — transfer to per-wheel (μ , χ) recovery is established only by analogy.
- (2) To our knowledge, **no prior work applies a selective state-space (Mamba) backbone** to Mecanum or mobile-robot forward-dynamics learning.

THIS PAPER'S CONTRIBUTION

A physics-informed, **Mamba-backed digital twin** that:

- Learns forward dynamics at high fidelity for a KUKA youBot Mecanum platform.
- Recovers interpretable per-wheel friction parameters (μ , χ) with physical bounds.
- Targets the **slip / friction / CoG chokepoint** that constrains Mecanum control fidelity.
- Bridges deployed kinematic control and the disturbance / slip / friction reality.

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